

E9: 309 ADL 02-12-2020



Housekeeping

Mock presentation

* Mid-term exam

→ December 6th (Sunday) [Topics covered up to Dec 2nd]

→ Mode of exam

First_LastName - Midterm1 . pdf

✓ Time to respond - 3 hours

- ◉ Exam paper uploaded in Teams Channel and response (photo-scanned and uploaded in your folder).

- ◉ Open book, open notes

- ★ Strictly no online communication or help sought.

- ★ Academic integrity and ethics strongly followed.



Recap of previous class



Variational auto encoders

✳ The data $\mathbf{x}$ and latent variable $\mathbf{z}$

✳ The forward model

- ✓ Sample the latent variable

- ✓ Sample the data given the latent $p_{\theta}(\mathbf{x}|\mathbf{z})$

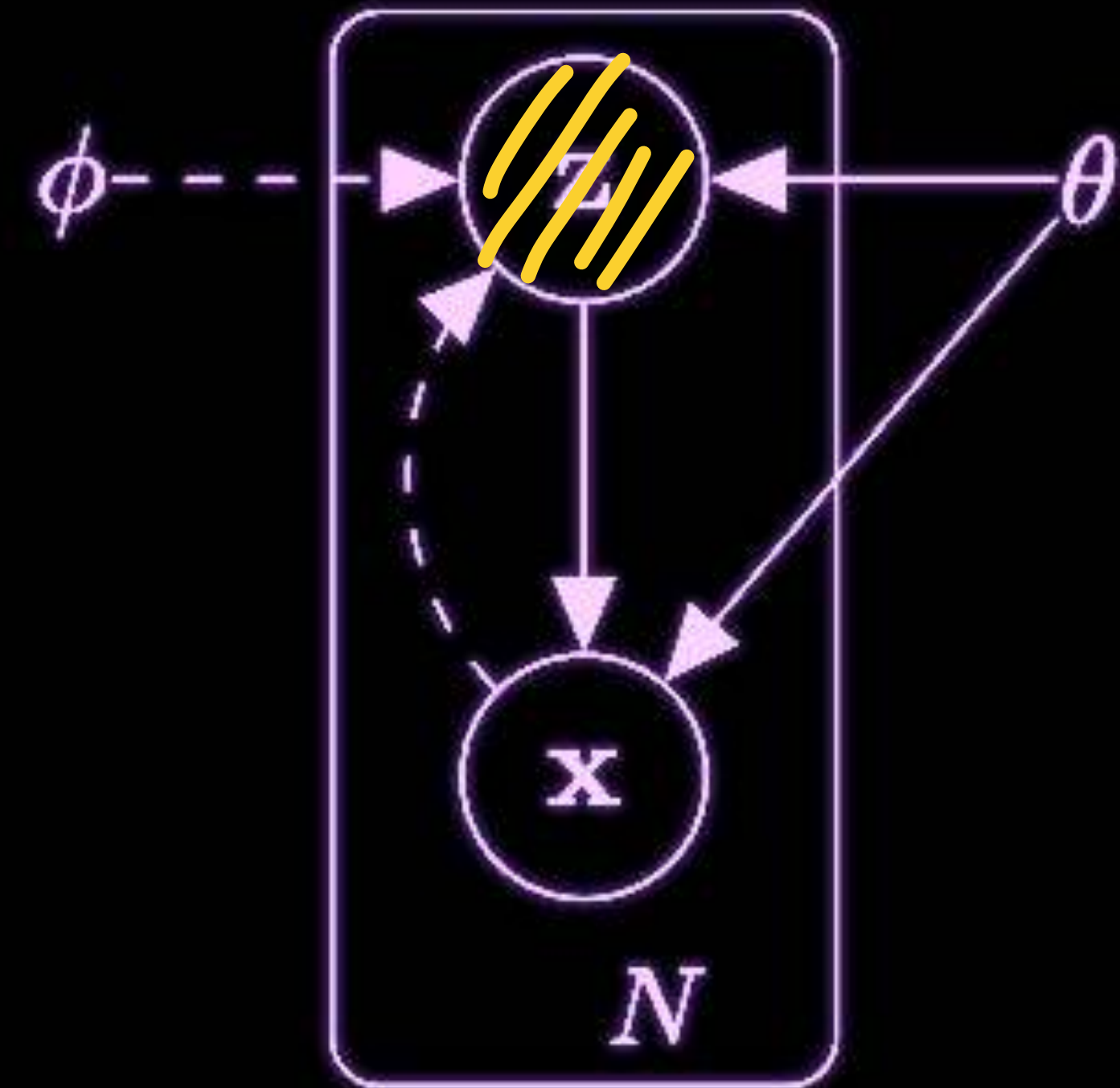
✳ The marginal distribution

- ✓ maybe intractable

$$\int p_{\theta}(\mathbf{x}|\mathbf{z})p_{\theta}(\mathbf{z})d\mathbf{z}$$

✳ The posterior distribution $p_{\theta}(\mathbf{z}|\mathbf{x})$ ←

➡ may also be intractable



Variational lower bound

* Defining the variational lower bound

$$L(\mathbf{x}; \phi, \theta) = \int q_\phi(\mathbf{z}|\mathbf{x}) \log \frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} d\mathbf{z} = \mathbb{E}_q \left(\log \left[\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z}|\mathbf{x})} \right] \right)$$

✓ Called ELBO (evidence lower bound of the data)

$$L(\mathbf{x}; \phi, \theta) = \log p_\theta(\mathbf{x}) - KL(q_\phi(\mathbf{z}|\mathbf{x}) || p_\theta(\mathbf{z}|\mathbf{x}))$$

$L = \mathbb{E}_{\mathbf{z} \sim q} (\log p_\theta(\mathbf{x}|\mathbf{z})) - KL(q_\phi(\mathbf{z}|\mathbf{x}) || p_\theta(\mathbf{z}|\mathbf{x}))$

$p_\theta(\mathbf{x}|\mathbf{z}) \sim \mathcal{N}(\mathbf{x}; f_\theta(\mathbf{z}), \mathbf{I})$

◉ Maximizing the lower bound

★ Maximizes the log likelihood of the data

★ Minimized the KL divergence between the true posterior distribution and approximation.



Reparameterization

- * Take a random variable independent of the data and parameters

$$\underline{\epsilon \sim p(\epsilon)}$$

- * Transform the random variable to the latent variable with parameters

$$\underline{\mathbf{z}} = g(\underset{\uparrow}{\epsilon}, \underset{\uparrow}{\phi}, \mathbf{x})$$

- * Now the gradient computation of the expectation can be simplified.

- ✓ Decoupling the sampling and the gradient computations on the term involving $\mathbb{E}_q \left(\log p_{\theta}(\mathbf{x}|\mathbf{z}) \right)$

$$\mathbb{E}_q \left(\log p_{\theta}(\mathbf{x}|\mathbf{z}) \right)$$



Assumptions and Approximations

$$L(\mathbf{x}; \phi, \theta) = \cancel{KL(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(z))} + \cancel{\mathbb{E}_q \left(\log p_\theta(\mathbf{x}|\mathbf{z}) \right)}$$

✱ Assume

$$p_\theta(z) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I}) \checkmark$$

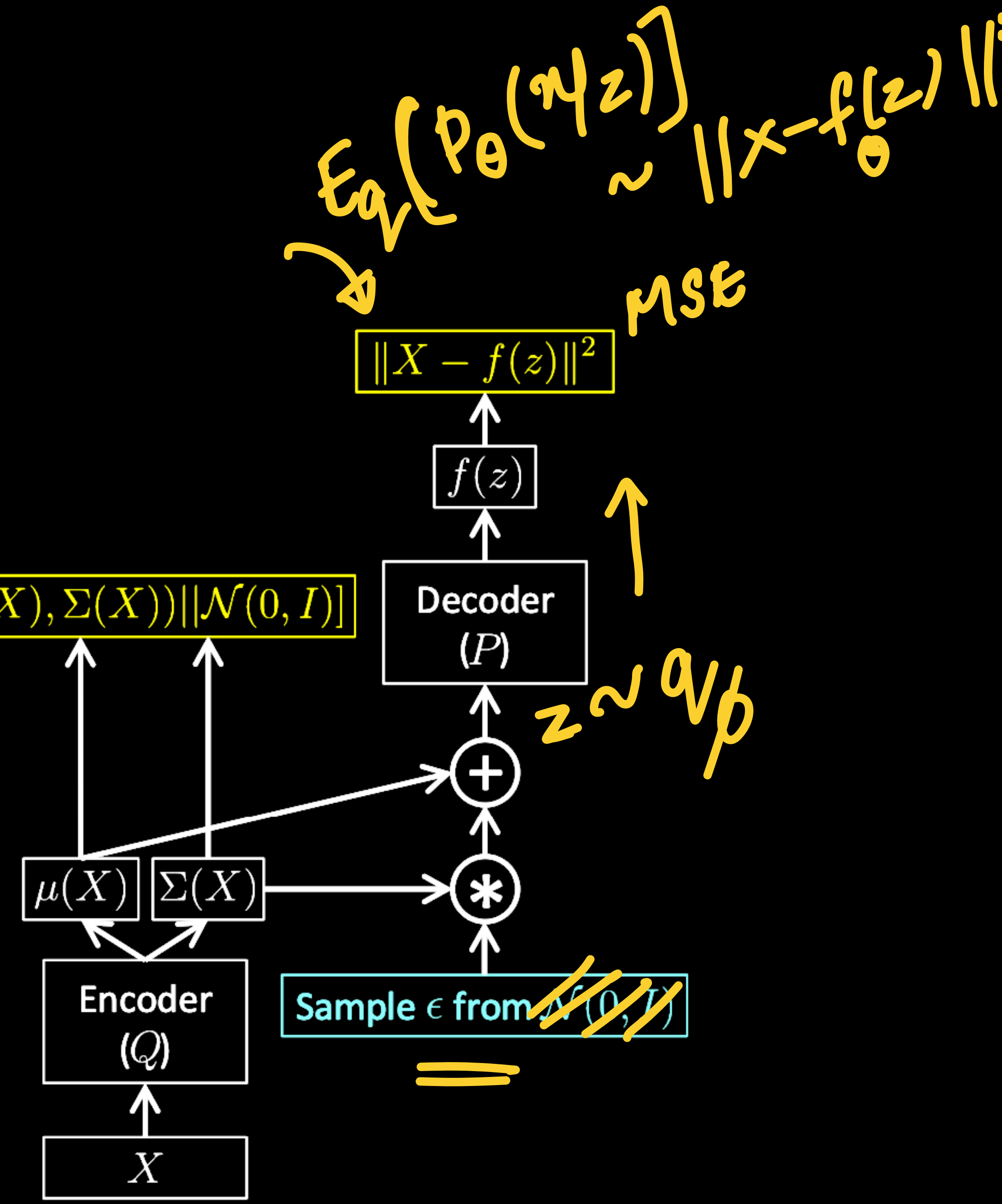
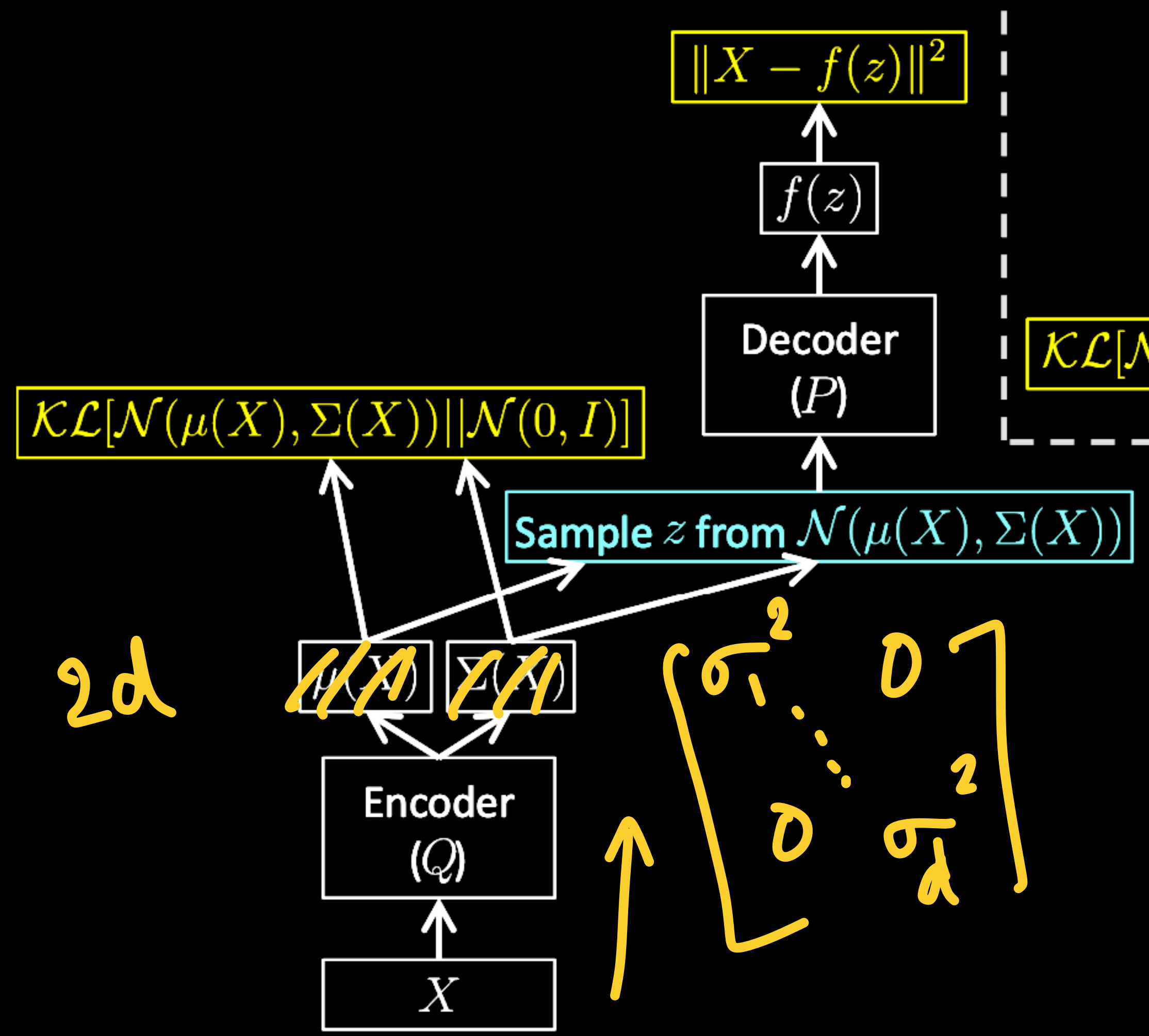
$$q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}(\mathbf{x}), \boldsymbol{\sigma}(\mathbf{x})^2 \mathbf{I}) \checkmark$$

➡ The conditional distribution

$$p_\theta(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}; \underbrace{f_\theta(\mathbf{z})}_{\theta}, \mathbf{I}) \checkmark$$

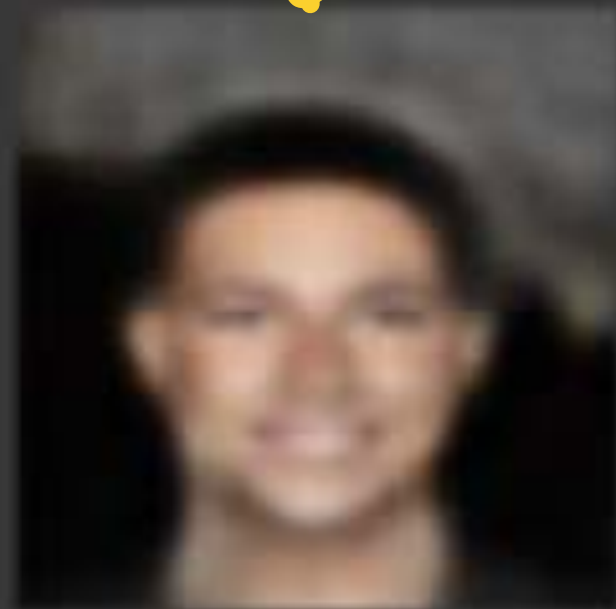


The algorithm



Standard AE - reconstruction

Celeb Faces Dataset



VAE = Regularized AE

$$\underbrace{KL(q_{\phi}(z|x) || p_{\theta}(z))}_{KL} \underbrace{|| x - f_{\theta}(z) ||^2}_{MSE}$$



Standard AE with added noise - reconstruction

Celeb Faces Dataset

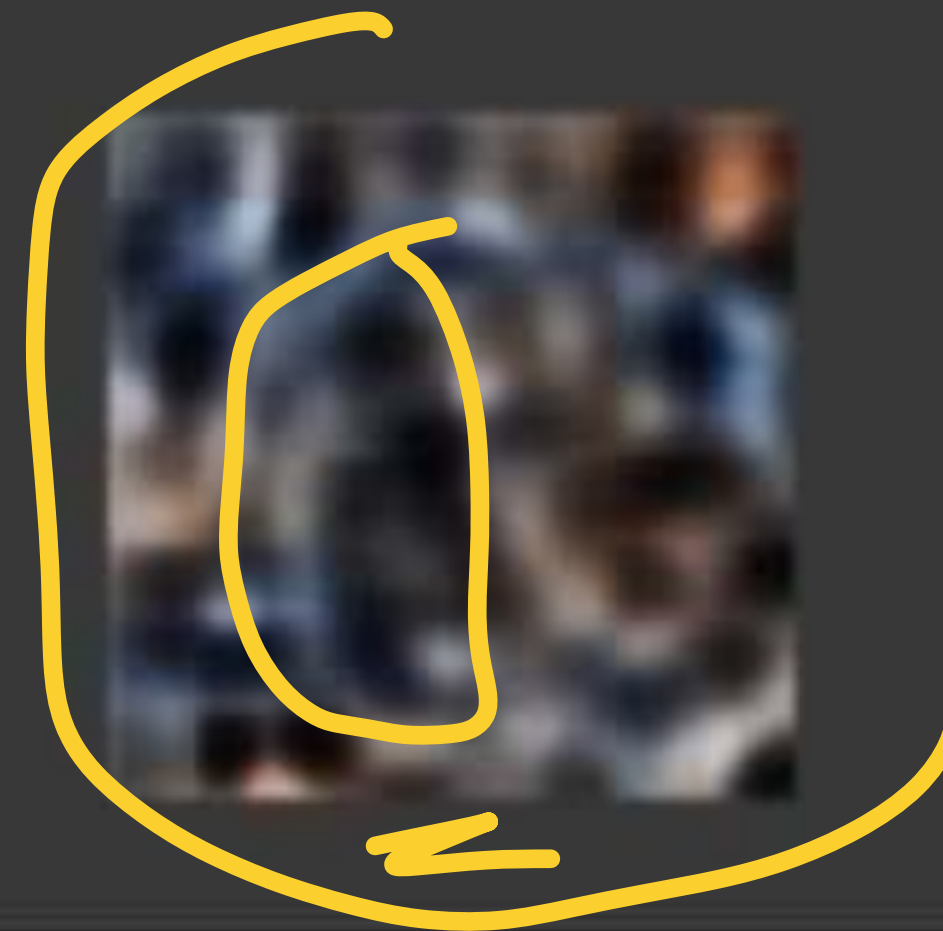
$x \xrightarrow{\text{Encoder}} z \quad (z = z + \Delta) \quad z \xrightarrow{\text{Decoder}} x'$



Standard AE only noise - reconstruction

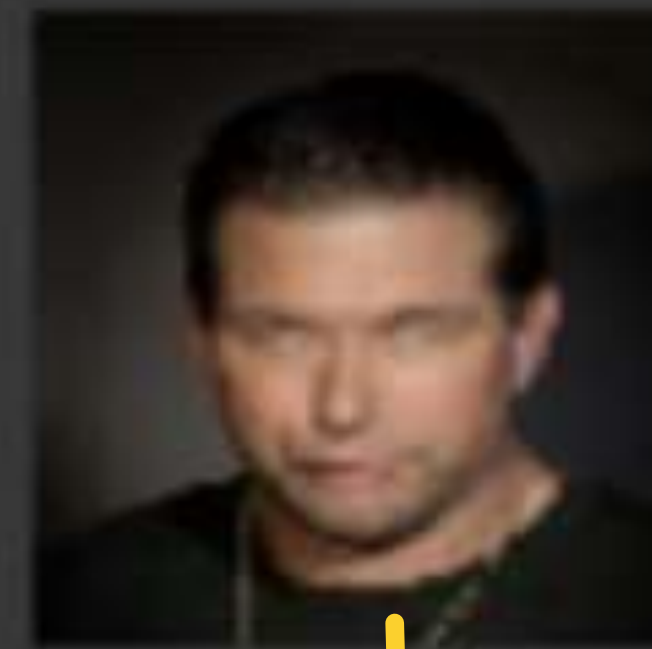
Δ *Decoder* $\rightarrow$

Using noise as latent vector



Variational Autoencoder - Using the sample

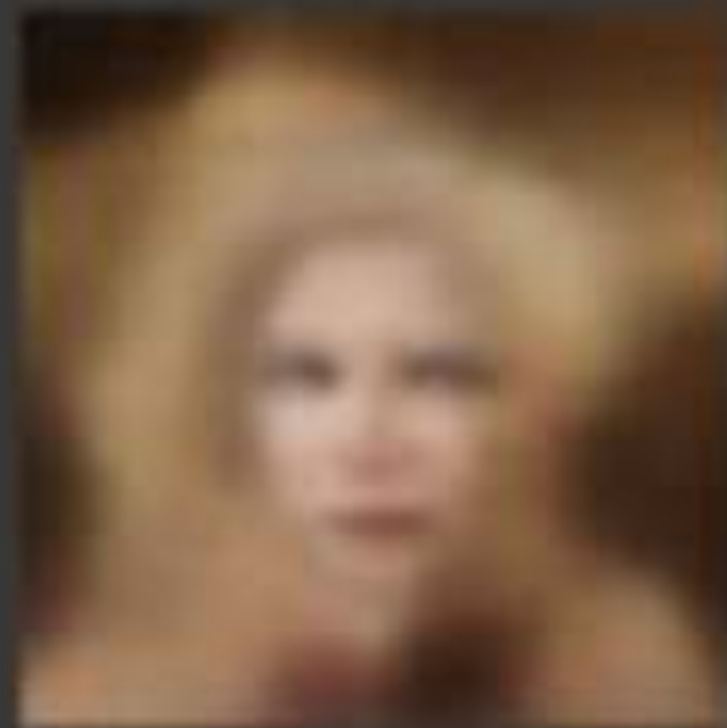
mode $\mu, \Sigma \leftarrow Z \sim N(\mu, \Sigma)$



Variational Autoencoder - Using the random noise

Using random noise generated from standard Gaussian - latent dim (200D)

quality of images can be improved



VAE

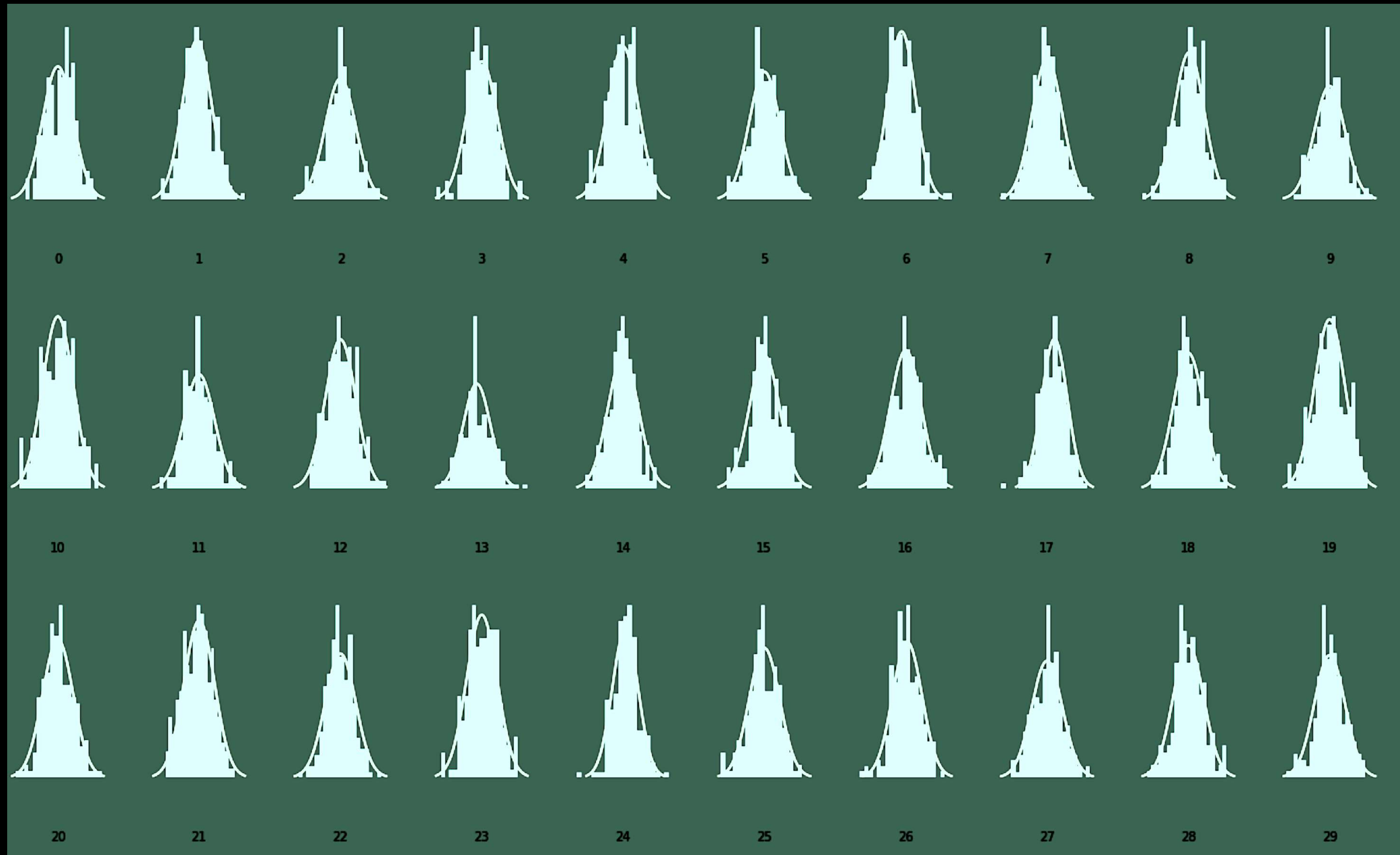
MNIST dataset

- ➡ linearly sampling from 2-D latent space.
- ➡ The latent space tends to cluster
 - ✓ Similar digits can be found by minor variations
 - ✓ There can be non-digits also in the output



VAE - latent distribution

Gaussian



Variational auto encoder

Kingma, Diederik P., and Max Welling. "Auto-encoding variational bayes." *arXiv preprint arXiv:1312.6114* (2013). ✓

✳ more Reading

→ Kingma, Diederik P., and Max Welling. "An introduction to variational autoencoders." *arXiv preprint arXiv:1906.02691* (2019).

✳ More reading

→ Doersch, Carl. "Tutorial on variational autoencoders." *arXiv preprint arXiv:1606.05908* (2016). CMU



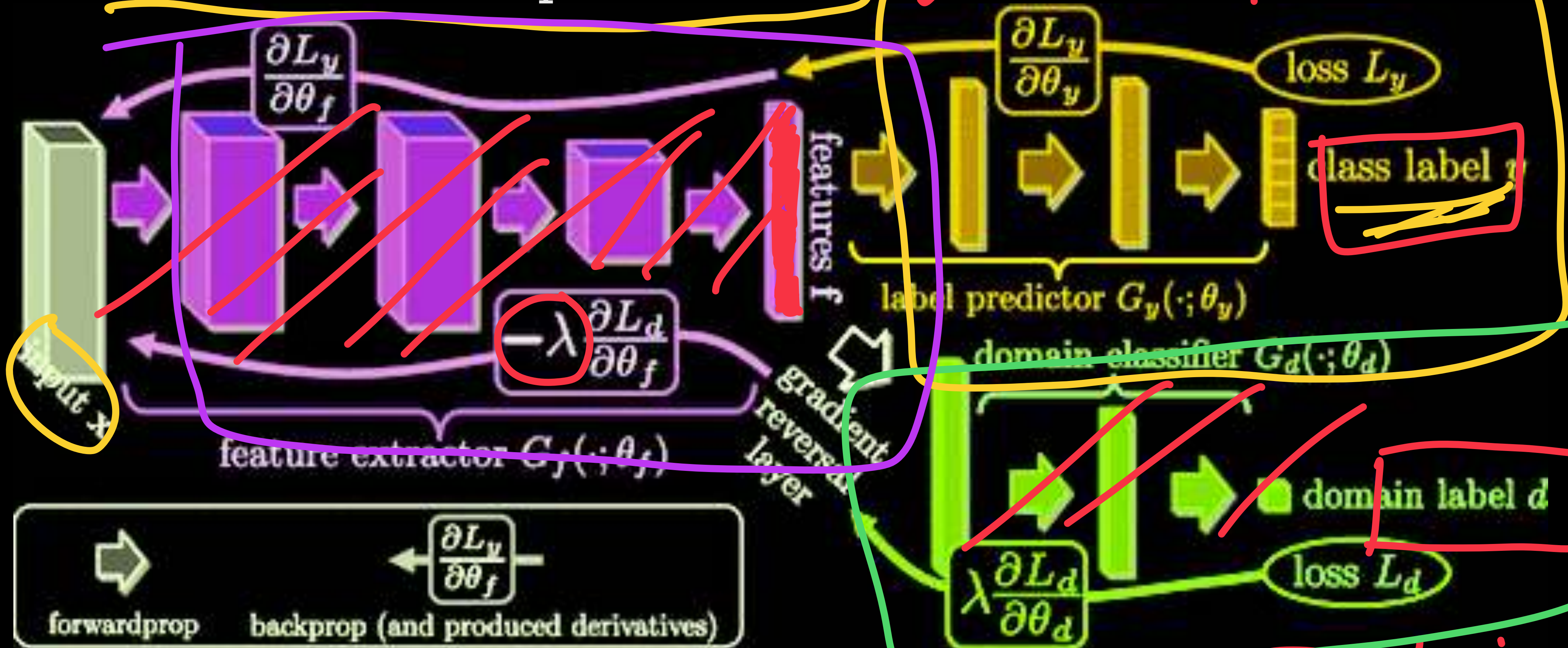
Adversarial loss [Supervised network]

Indoor
Outdoor

✳ Increasing the loss on certain nuisance directions

➔ To increase the invariance of the representations

Domain
Adversarial
Loss



Generative Adversarial Networks (GAN)

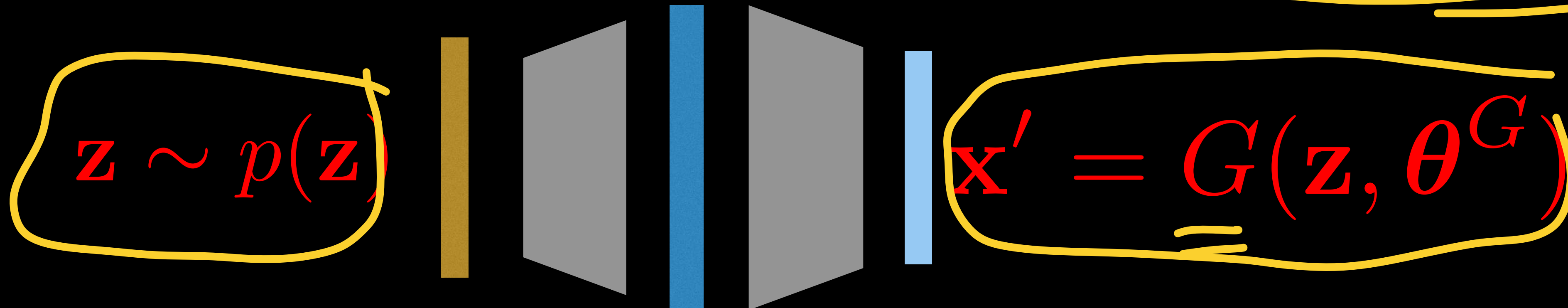
* Generator networks ✓

→ Differentiable network

→ Draws a random noise sample $\sim p(z)$ [which is parameter]

✓ Convert the noise to the data distribution

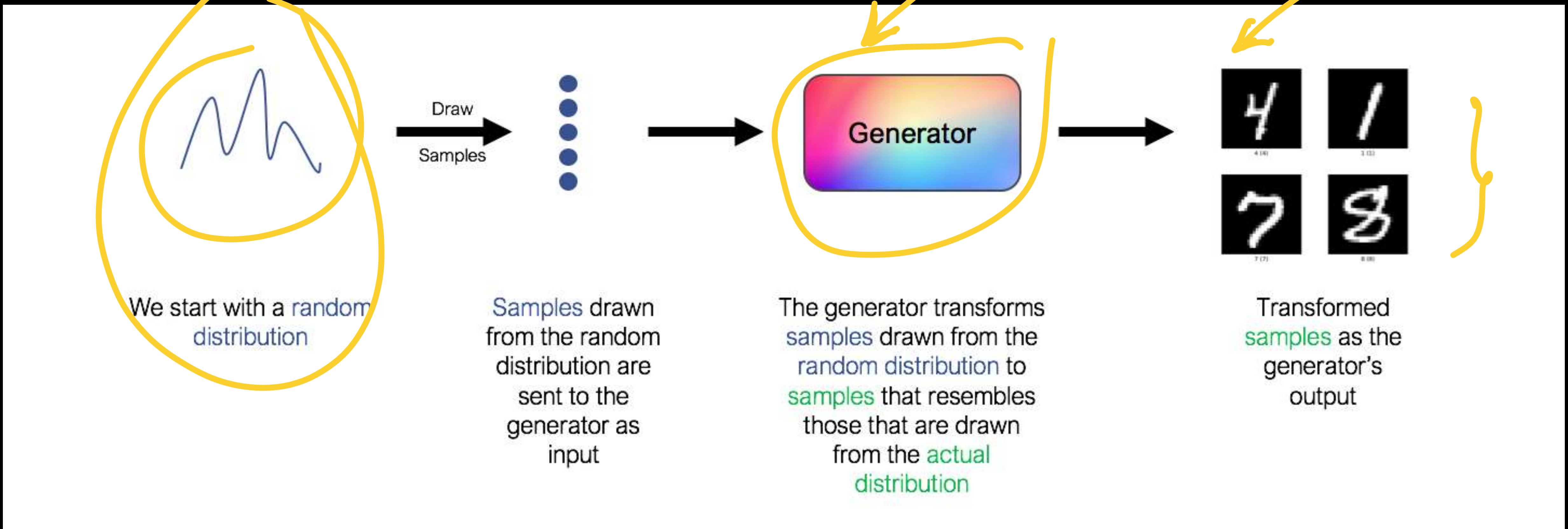
$$\mathbf{x}' \sim p_{\theta^G}(\mathbf{x}|\mathbf{z})$$



GAN - generator

$$x \sim p_x(x)$$
$$G(x) \sim p_{data}$$

learn unknown data distn



Generative Adversarial Networks

* Discriminator network

→ Two class classifier

✓ Label 0 for x'

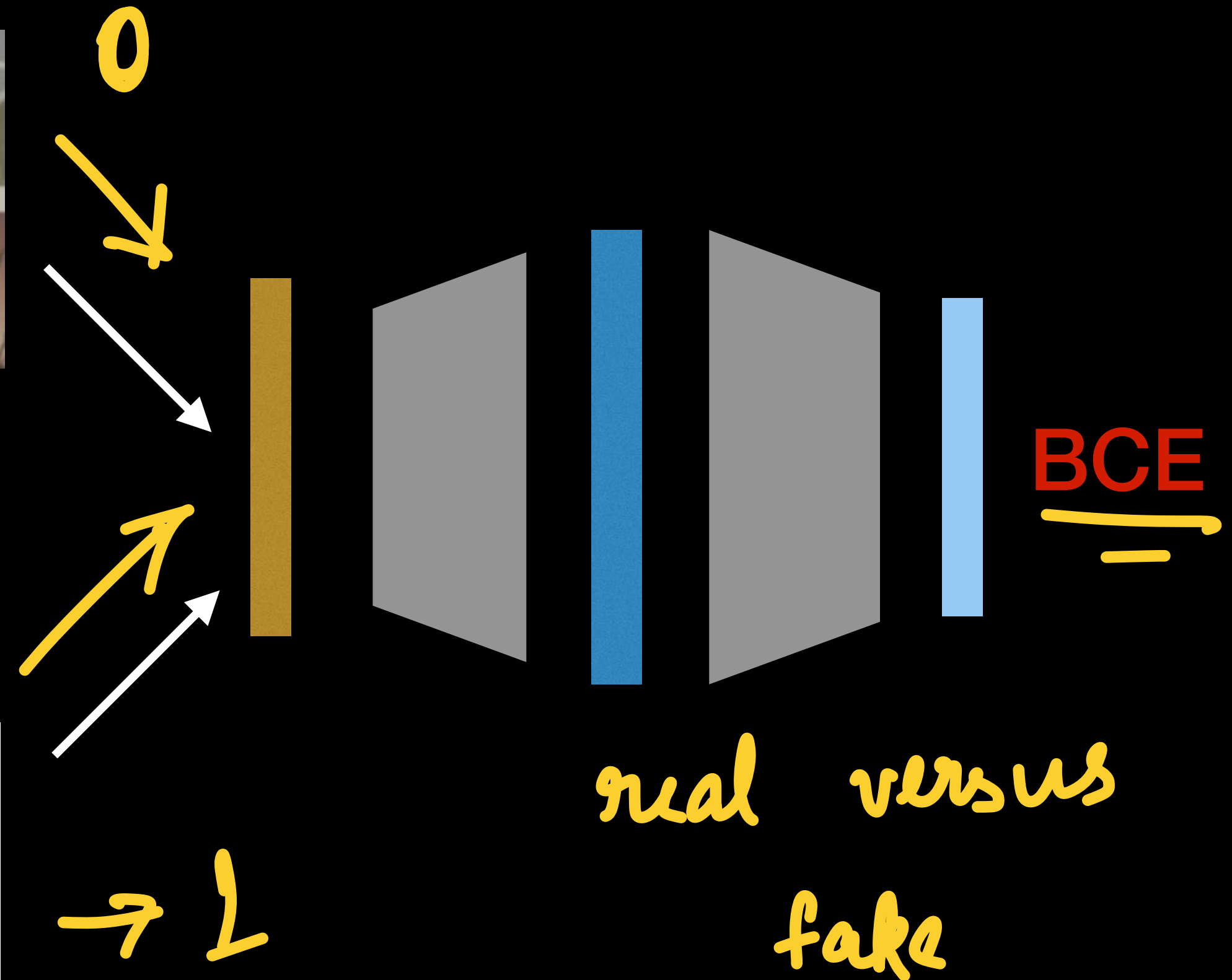
✓ Label 1 for (x)

→ Model learning

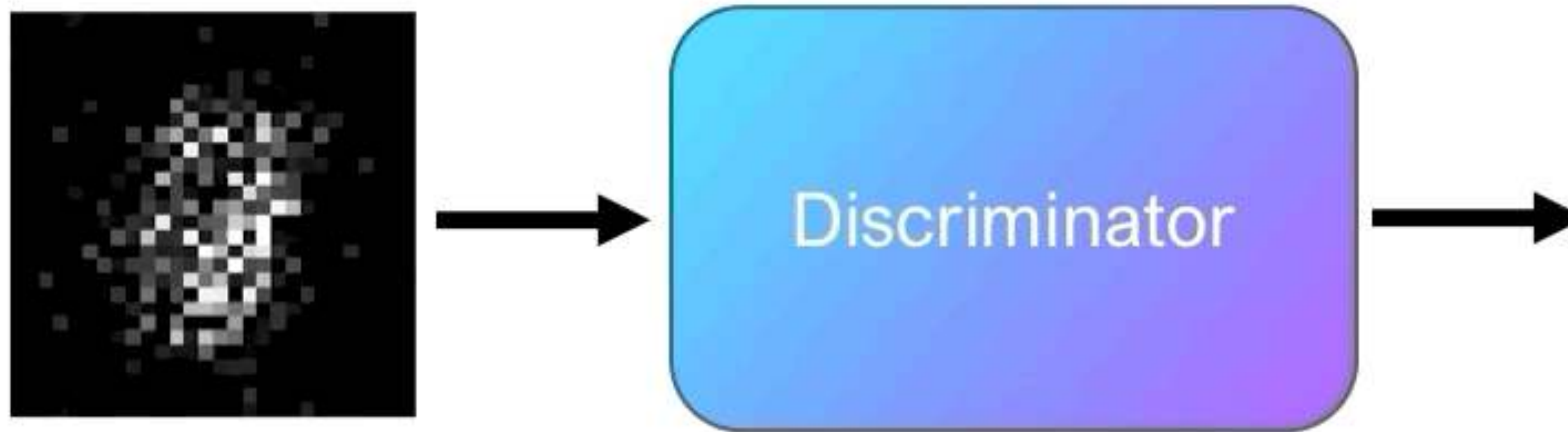
Model generated samples



True samples



GAN - discriminator



The discriminator takes images as input, and predicts the most likely class (real or fake) of the input image

Min-max game

✳ The binary cross entropy loss at the discriminator network is

$$\log [1 - \underbrace{D(G(z))}_{x']$$

$$E_D = -\mathbb{E}_{\mathbf{x} \sim p_{data}(\mathbf{x})} [\log D(\mathbf{x})] - \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} [\log D(G(\mathbf{z}))]$$

$$E_G = -E_D$$

$$- \underbrace{t_n}_{1} \log y_n - \underbrace{(1-t_n)}_0 \log (1-y_n)$$

✳ The discriminator is trained using B.C.E loss

$$\theta_D^n = \theta_D^{n-1} - \eta \frac{\partial E_D}{\partial \theta_D}$$

θ_D - discriminator
NN
parameters

η - iteration



Min-max game

✳ The binary cross entropy loss at the discriminator network is

$$E_D = -\mathbb{E}_{\mathbf{x} \sim p_{data}(\mathbf{x})} [\log D(\mathbf{x})] - \mathbb{E}_{\mathbf{z} \sim p(\mathbf{z})} [\log D(G(\mathbf{z}))]$$

$$E_G = -E_D$$

will not affect G.

change the generator

$$\log(1 - D(G(\mathbf{z})))$$

✳ The discriminator is trained using B.C.E loss

Generator is trying to defeat the D.

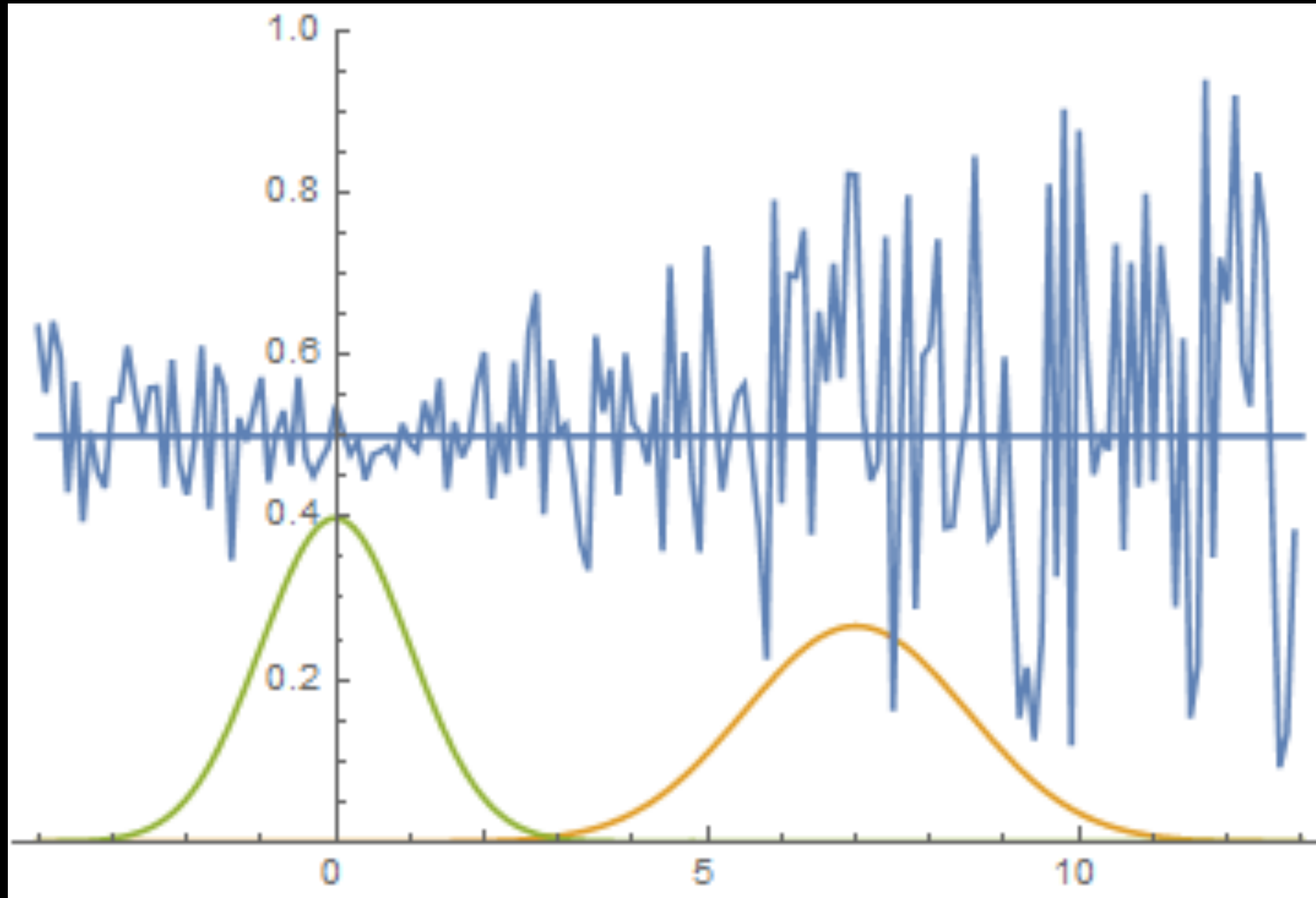
$$\theta_G^n = \theta_G^{n-1} - \eta \frac{\partial E_G}{\partial \theta_G}$$



GAN - Simple example

Gaussian
on
left is
distⁿ of model
generated
outputs

plot is
output
of discrim
minator



Gaussian
on
the right
- true
data
distrⁿ

GAN - considerations

- Usually the discriminator “wins”
 - This is a good thing—the theoretical justifications are based on assuming D is perfect
- Usually D is bigger and deeper than G
 - The D starts off with much faster saturation.
- Learning of G happens even when D is saturated
- Equilibrium - Discriminator is defeated and outputs half for both real and model generated samples.

